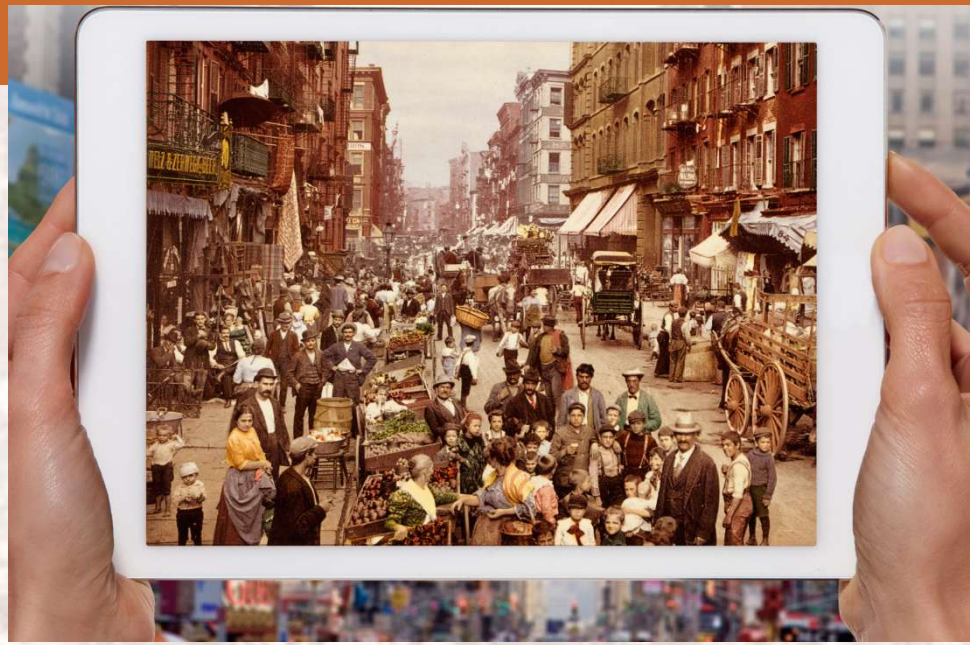
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CHAPTER

5

Elasticity and Its Application

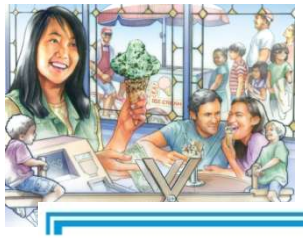
National Central University

Look for the answers to these questions:

- What is elasticity?
- What kinds of issues can elasticity help us understand?
- What is the price elasticity of demand?
  How is it related to the demand curve?
  How is it related to revenue & expenditure?
- What is the price elasticity of supply?
  How is it related to the supply curve?
- What are the income and cross-price elasticities of demand (需求的交叉彈性)?

A scenario:

- You design websites for local businesses.
  - You charge $200 per website, and currently sell 12 websites per month.
- Your costs are rising (including the opportunity cost of your time)
  - You consider raising the price to $250.
- The law of demand: you won't sell as many websites if you raise your price.
  - How many fewer websites?
  - How much will your revenue fall, or might it increase?



The Elasticity of Demand

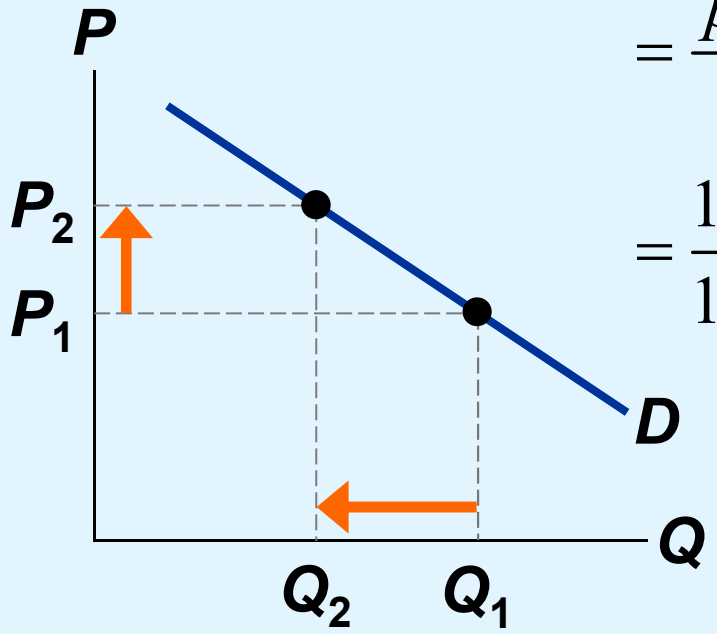
- Elasticity
  - Measure of the responsiveness of Q^d or Q^s
    - To a change in one of its determinants
- Price elasticity of demand
  - How much the quantity demanded of a good responds to a change in the price of that good
    - Loosely speaking, it measures the price-sensitivity of buyers' demand

Price Elasticity of Demand $E = \left| \frac{\%\ \text{change in quantity demand}}{\%\ \text{change in price}} \right| = \left| \frac{p}{q} \cdot \frac{dq}{dp} \right|$

Price elasticity of demand =

$$= \frac{percentage \quad change \quad in \quad Q^d}{percentage \quad change \quad in \quad P}$$

$$= \frac{15\%}{10\%} = 1.5$$



P rises by 10%

Q falls by 15%

Along a D curve, P and Q move in opposite directions, which would make ***price elasticity negative***.

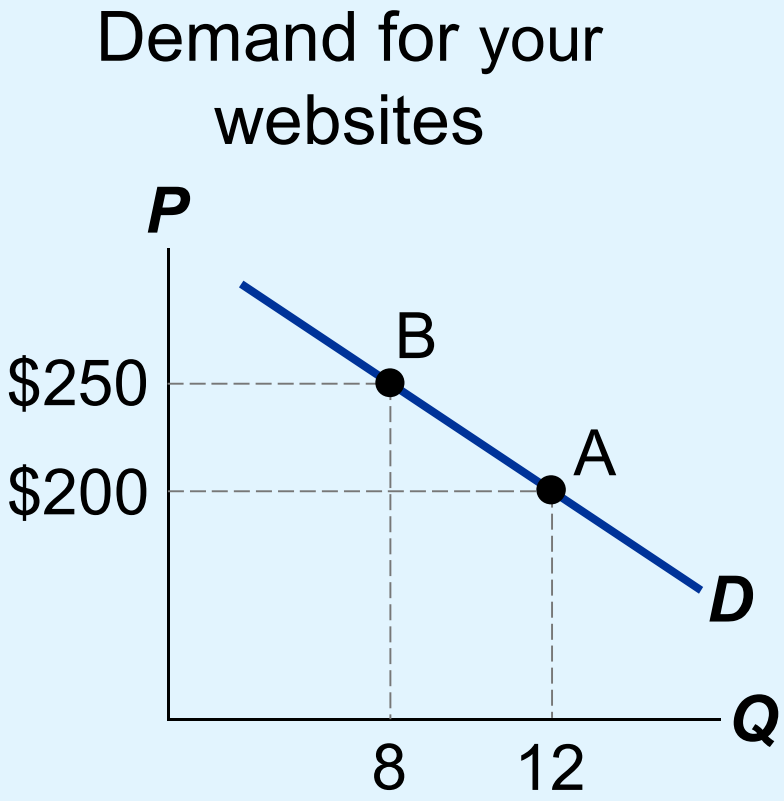
We will drop the minus sign and report all price elasticities as positive numbers.

Question

For each of the following pairs of goods, which good would you expect to have more elastic demand?

a. required textbooks or mystery novels
b. Beethoven recordings or classical music recordings in general
c. Subway rides during the next 6 months or subway rides during the next 5 years
d. Root beer or water

Calculating Percentage Changes



Standard method of computing the percentage (%) change:

$$= \frac{end \quad value - start \quad value}{start \quad value} \times 100\%$$

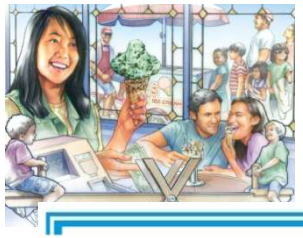
Going from A to B:

- the % change in P = ($250–$200)/$200 = 25%
- the % change in Q = (8-12)/12 = - 33%
- Price elasticity = 33/25 = **1.32**

Going from B to A:

- the % change in P = ($200-$250)/$250 = - 20%
- the % change in Q = (12-8)/8 = 50%
- Price elasticity = 50/20 = **2.5**

We get different values!



The Price Elasticity of Demand

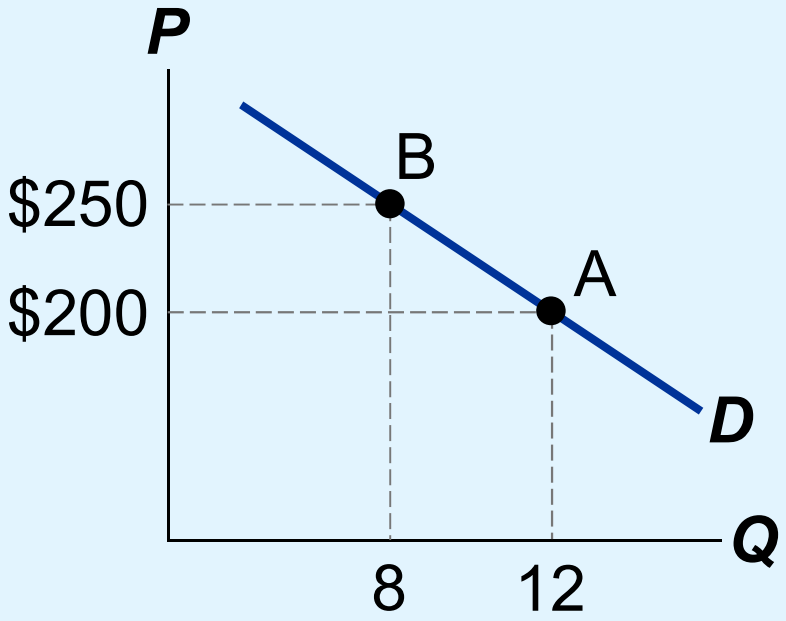
- Midpoint method
  - The midpoint is the number halfway between the start and end values
    - The average of those values

$$\text{percentage change} = \frac{\text{end value - start value}}{\text{midpoint}} \times 100\%$$

$$\text{Price elasticity of demand} = \frac{(Q_2 - Q_1) / [(Q_2 + Q_1) / 2]}{(P_2 - P_1) / [(P_2 + P_1) / 2]}$$

Calculating Percentage Changes

Demand for your websites



Using the midpoint method of computing % changes:

$$\%\ \text{change in P} = \frac{\$250 - \$200}{\$225} \times 100\% = 22.2\%$$

$$\%\ \text{change in Q} = \frac{12 - 8}{10} \times 100\% = 40\%$$

$$\text{Price elasticity} = \frac{40\%}{22.2\%} = 1.8$$

Active Learning 1 — Calculate an elasticity

Use the following information to calculate the price elasticity of demand for iPhones:

- if P = \$400, Q^d = 10,600
- if P = \$600, Q^d = 8,400
- Use the midpoint method to calculate percentage changes.

Active Learning 1 Answers

Using the midpoint method to calculate percentage changes:

- % change in P =

  [($600 - $400)/$500] ×100 = 40%

- % change in Q^d =

  [(10,600 – 8,400)/9,500] ×100 = 23.16%

- Price elasticity of demand =

  = % change in Q^d / % change in P

  = 23.16/40 = 0.58

Question

Suppose that business travelers and vacationers have the following demand for airline tickets from New York to Boston:

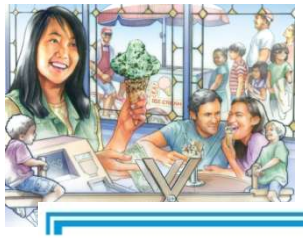
Price	Quantity Demanded (business travelers)	Quantity Demanded (vacationers)
$150	2,100 tickets	1,000 tickets
200	**2,000**	**800**
250	**1,900**	**600**
300	1,800	400

a. As the price of tickets rises from $200 to $250, what is the price elasticity of demand for (i) business travelers and (ii) vacationers? (Use the midpoint method in your calculations.)
b. Why might vacationers have a higher elasticity from business travelers?

Question

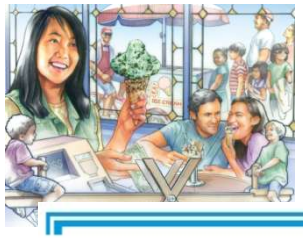
Suppose the price elasticity of demand for heating oil is 0.2 in the short run and 0.7 in the long run

a. If the price of heating oil rises from $1.80 to $2.20 per gallon, what happens to the quantity of heating oil demanded in the short run? In the long run? (Use the midpoint method in your calculations)
b. Why might this elasticity depend on the time horizon?



The Price Elasticity of Demand

- Determinants of price elasticity of demand
  - We look at a series of examples comparing two common goods
- In each example:
  - Suppose prices of both goods rise by 20%
  - Which good has the highest price elasticity of demand? Why?
  - What lesson we learn about the determinants of price elasticity of demand?

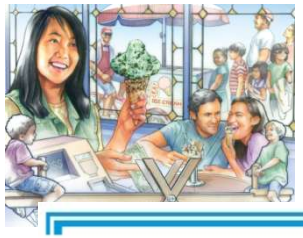


The Price Elasticity of Demand

Example 1: Breakfast cereal vs. Sunscreen

- Prices of both of these goods rise by 20%. For which good does Q^d drop the most? Why?
  - Breakfast cereal has close substitutes, so buyers can easily switch if the price rises
  - Sunscreen has no close substitutes, so a price increase would not affect demand very much

- Price elasticity is higher when close substitutes are available

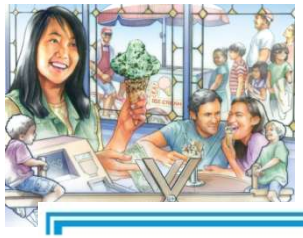


The Price Elasticity of Demand

Example 2: Blue Jeans vs. Clothing

- Prices of both of these goods rise by 20%. For which good does Q^d drop the most? Why?
  - For a narrowly defined good, blue jeans, there are many substitutes
    There are fewer substitutes available for broadly defined goods (clothing)

Price elasticity is higher for narrowly defined goods than for broadly defined ones.

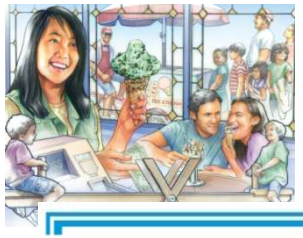


The Price Elasticity of Demand

Example 3: Insulin vs. Yachts

- Prices of both of these goods rise by 20%. For which good does Q^d drop the most? Why?
  - Insulin is a necessity to diabetics. A rise in price would cause little or no decrease in demand
  - A yacht is a luxury. If the price rises, some people will forego it.

- Price elasticity is higher for luxuries than for necessities.

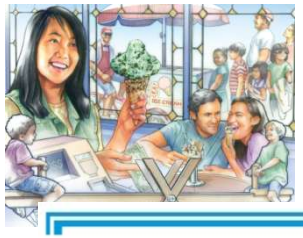


The Price Elasticity of Demand

Example 4: Gasoline in the Short Run vs. Gasoline in the Long Run

- The price of gasoline rises 20%. Does Q^d drop more in the short run or the long run? Why?
  - There's not much people can do in the short run, other than ride the bus or carpool.
  - In the long run, people can buy smaller cars or live closer to work.

- Price elasticity is higher in the long run



The Price Elasticity of Demand

- Variety of demand curves
  - Demand is elastic
    - Price elasticity of demand > 1
  - Demand is inelastic
    - Price elasticity of demand < 1
  - Demand has unit elasticity
    - Price elasticity of demand = 1

$$E_d = -\frac{\partial Q}{\partial P} \cdot \frac{P}{Q}$$

當$E_d = 0$：完全無彈性
當$0 < E_d < 1$：缺乏彈性
當$E_d = 1$：單位彈性
當$1 < E_d < \infty$：富有彈性
當$E_d = \infty$ ：完全有彈性

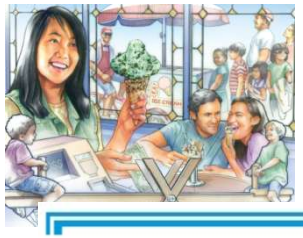
Question

A price change causes the quantity demanded of a good to decrease by 30 percent, while the total revenue of that good increases by 15 percent. Is the demand curve elastic or inelastic? Explain.

Key Question

- Cups of coffee and donuts are complements
- Both have inelastic demand
- A hurricane destroys half the coffee bean crop

a. What happens to the price of coffee beans?
b. What happens to the price of a cup of coffee? What happens to total expenditure on cups of coffee?
c. What happens to the price of donuts? What happens to total expenditure on donuts?

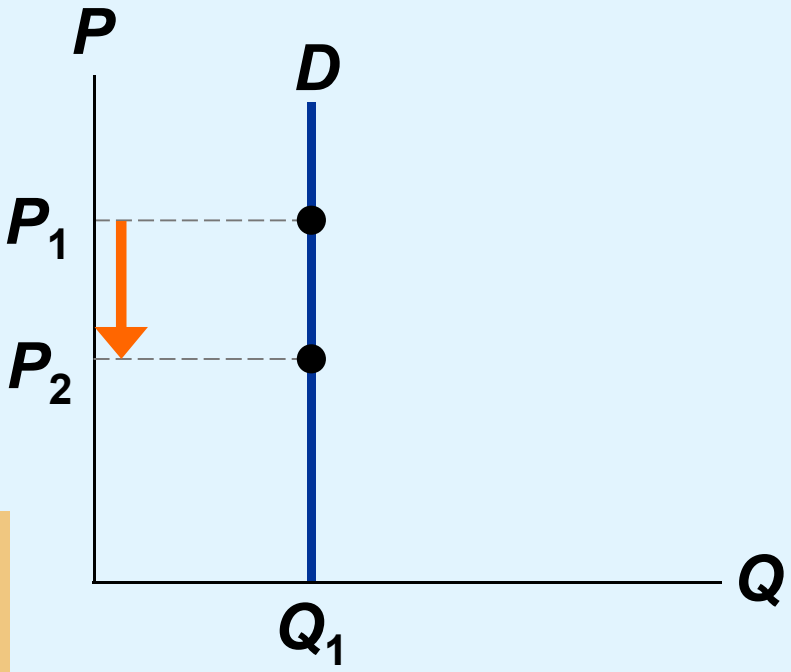


The Price Elasticity of Demand

- Variety of demand curves
  - Demand is perfectly inelastic
    - Price elasticity of demand = 0
    - Demand curve is vertical
  - Demand is perfectly elastic
    - Price elasticity of demand = infinity
    - Demand curve is horizontal
- The flatter the demand curve
  - The greater the price elasticity of demand

Perfectly inelastic demand

$$\text{Price elasticity of demand} = \frac{\%\text{ change in } \boldsymbol{Q}}{\%\text{ change in } \boldsymbol{P}} = \frac{0\%}{10\%} = 0$$



P falls by 10%

Q changes by 0%

D curve

Vertical

Consumers' price sensitivity:

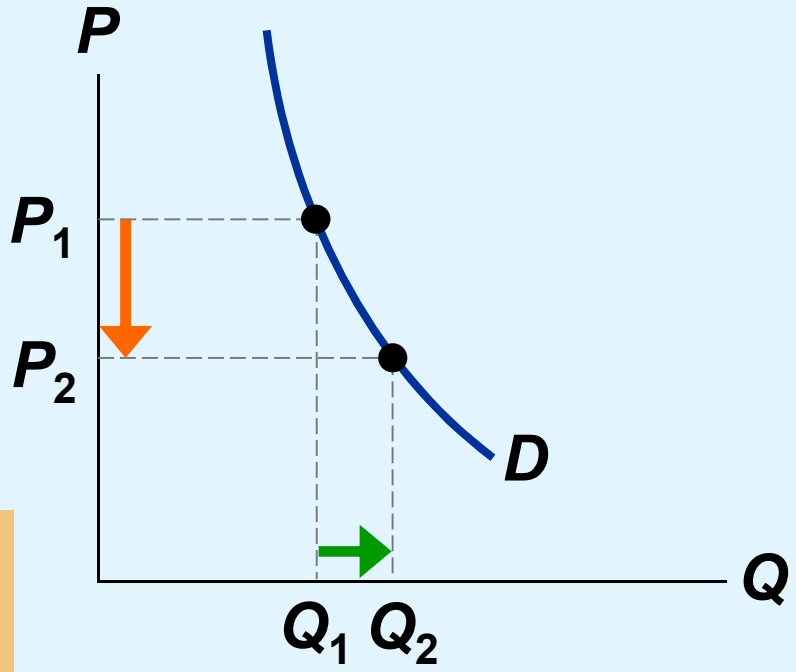
None

Elasticity:

0

Inelastic demand

$$\text{Price elasticity of demand} = \frac{\%\text{ change in } \boldsymbol{Q}}{\%\text{ change in } \boldsymbol{P}} = \frac{<10\%}{10\%} < 1$$



P falls by 10%

Q rises less than 10%

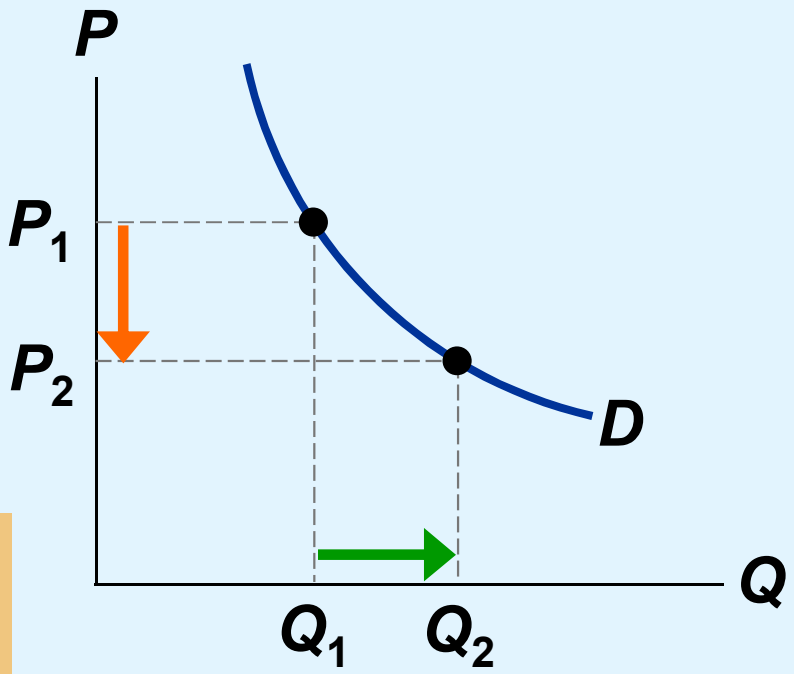
D curve
relatively steep

Consumers' price sensitivity:
relatively low

Elasticity:
<1

Unit elastic demand

$$\text{Price elasticity of demand} = \frac{\%\ \text{change in } \boldsymbol{Q}}{\%\ \text{change in } \boldsymbol{P}} = \frac{10\%}{10\%} = 1$$



P falls by 10%

Q rises by 10%

D curve

intermediate slope

Consumers' price sensitivity:

intermediate

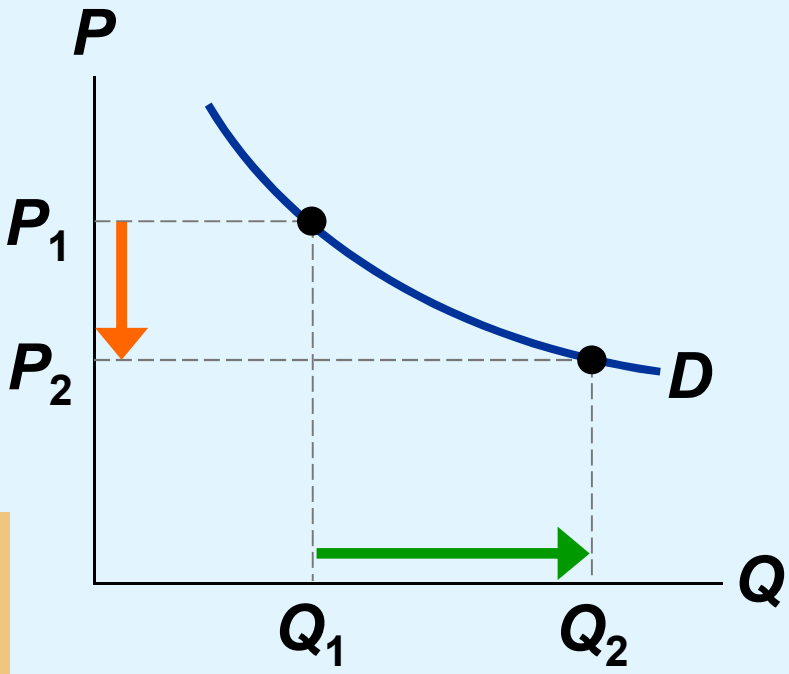
Elasticity:

=1

此時價格改變不會影響收益(Revenue)

Elastic demand

$$\text{Price elasticity of demand} = \frac{\%\text{ change in } \boldsymbol{Q}}{\%\text{ change in } \boldsymbol{P}} = \frac{>10\%}{10\%} > 1$$



P falls by 10%

Q rises more than 10%

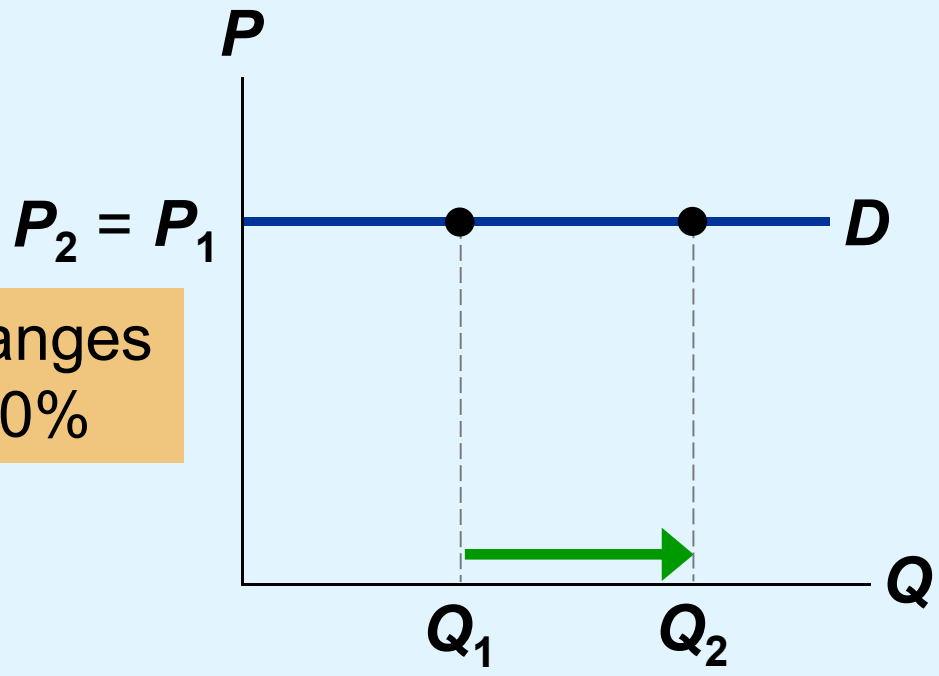
D curve relatively flat

Consumers' price sensitivity: relatively high

Elasticity: >1

Perfectly elastic demand

$$\text{Price elasticity of demand} = \frac{\%\ \text{change in } \boldsymbol{Q}}{\%\ \text{change in } \boldsymbol{P}} = \frac{\text{any } \%}{0\%} = \text{infinity}$$



P changes by 0%

Q changes by any %

D curve
horizontal

Consumers' price sensitivity:
extreme

Elasticity:
infinity

A Few Elasticities from the Real World

Eggs	0.1
Healthcare	0.2
Cigarettes	0.4
Rice	0.5
Housing	0.7
Beef	1.6
Peanut Butter	1.7
Restaurant Meals	2.3
Mountain Dew	4.4

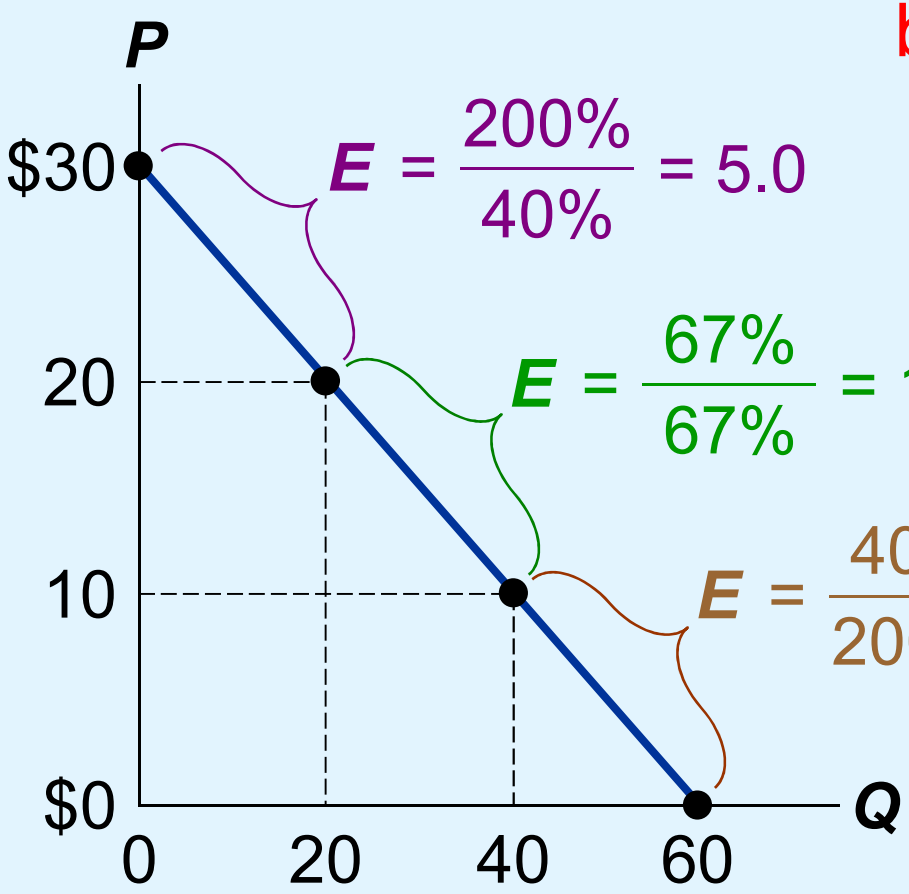


The reason Aldi's milk prices are unusually low

Elasticity along a Linear Demand Curve

最有彈性的地方是消費者買的量很少
但再多買一單位所願意付的價格很高

The slope of a linear demand curve is constant, but its elasticity is not.



最沒彈性的地方是消費者買的量很多
但再多買一單位所願意付的價格很低

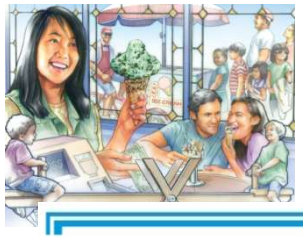
雖然斜率固定是一塊錢增加造成兩單位商品減少，但彈性計算方式是看數量變動百分比與價格變動百分比，因此目前數量與價格的大小會造成需求線段上的每一點的價格彈性不相同

Price Elasticity and Total Revenue

Continuing our scenario, if you raise your price from $200 to $250, would your revenue rise or fall?

Total Revenue (TR) = P x Q

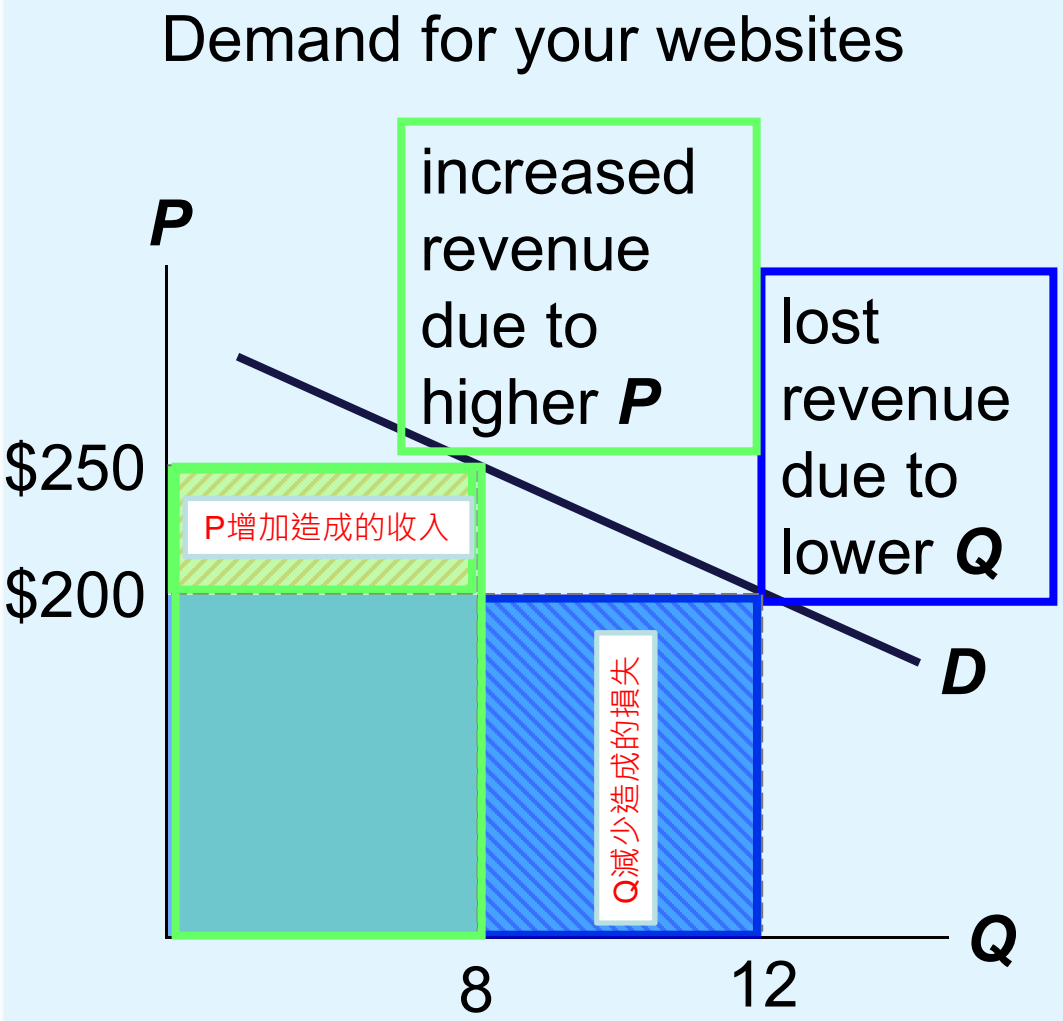
- A price increase has two effects on revenue:
  - Higher revenue: because of the higher P
  - Lower revenue: you sell fewer units (lower Q)
- Which of these two effects is bigger?
  - It depends on the price elasticity of demand



Price Elasticity and Total Revenue

- For a price increase, if demand is elastic
  - E > 1: % change in Q > % change in P
  - TR decreases: the fall in revenue from lower Q > the increase in revenue from higher P
- For a price increase, if demand is inelastic
  - E < 1: % change in Q < % change in P
  - TR increases: the fall in revenue from lower Q < the increase in revenue from higher P

Price Elasticity and Total Revenue



$$\frac{\frac{12-8}{(12+8)/2}}{\frac{250-200}{(250+200)/2}} = \frac{900}{500} = 1.8$$

Elastic demand (elasticity = 1.8)

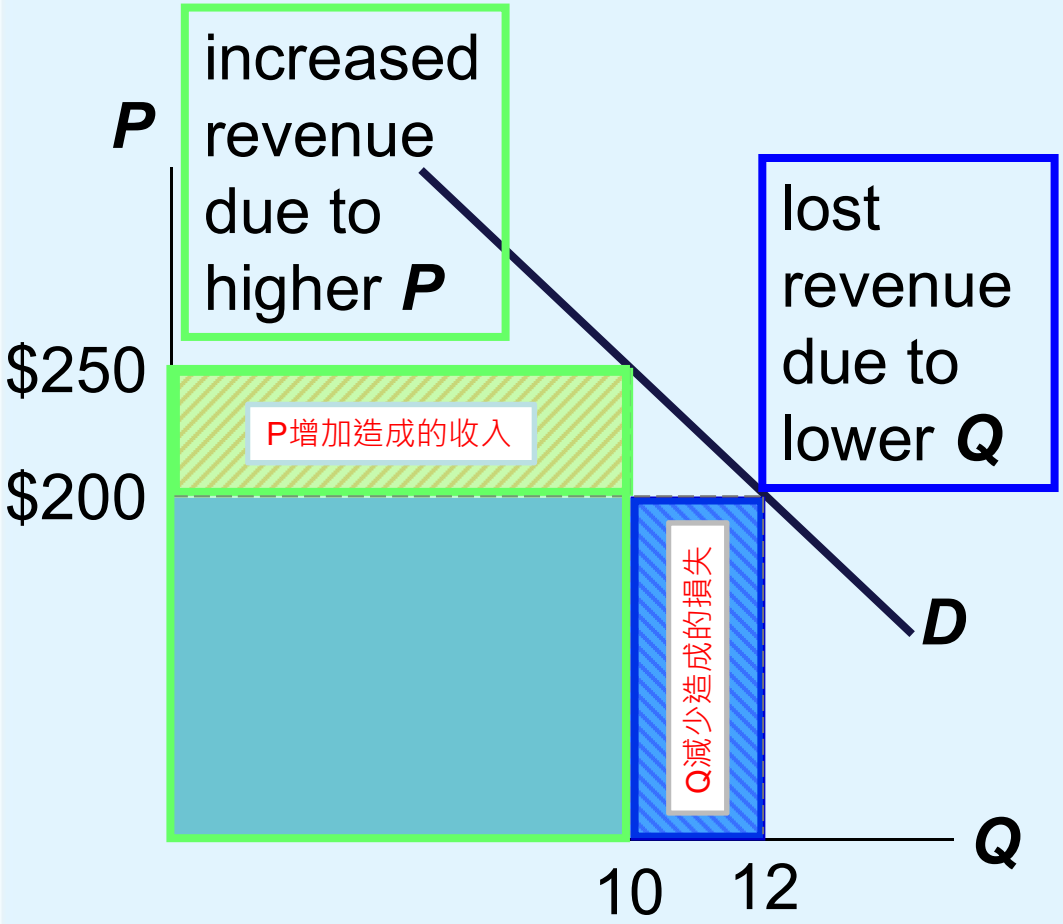
If ***P*** = \$200, ***Q*** = 12, and revenue = \$2400

If ***P*** = \$250, ***Q*** = 8, and revenue = \$2000

When ***D*** is elastic, a price increase causes revenue to fall.

Price Elasticity and Total Revenue

Demand for your websites



Inelastic demand (elasticity = 0.82)

If ***P*** = $200, ***Q*** = 12, and revenue = $2400

If ***P*** = $250, ***Q*** = 10, and revenue = $2500

When ***D*** is inelastic, a price increase causes revenue to rise.

Active Learning 2 Elasticity and revenue

A. Pharmacies raise the price of insulin by 10%.
   - Does total expenditure on insulin rise or fall?

B. As a result of a fare war, the price of a luxury cruise falls 20%.
   - Does luxury cruise companies' total revenue rise or fall?



The expenditure paid by consumers is the revenue of the manufacturer.

Active Learning 2 Answers

A. Pharmacies raise the price of insulin by 10%.

- Does total expenditure on insulin rise or fall?

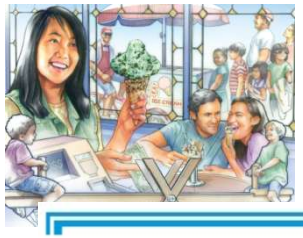
- Expenditure = P x Q
- Since demand is inelastic, Q will fall less than 10%, so expenditure rises.

Active Learning 2 Answers

B. As a result of a fare war, the price of a luxury cruise falls 20%.

– Does luxury cruise companies' total revenue rise or fall?

- Revenue = P x Q
- The fall in P reduces revenue, but Q increases, which increases revenue. Which effect is bigger?
- Since demand is elastic, Q will increase more than 20%, so revenue rises.



Does Drug Interdiction Increase or Decrease Drug-related Crime?

1. Increase the number of federal agents devoted to the war on drugs
   - Illegal drugs: supply curve shifts left
     - Higher price and lower quantity
   - Amount of drug-related crimes
     - Inelastic demand for drugs 毒品上癮造成需求缺乏彈性
     - Higher drugs price: higher total revenue
     - Increase drug-related crime

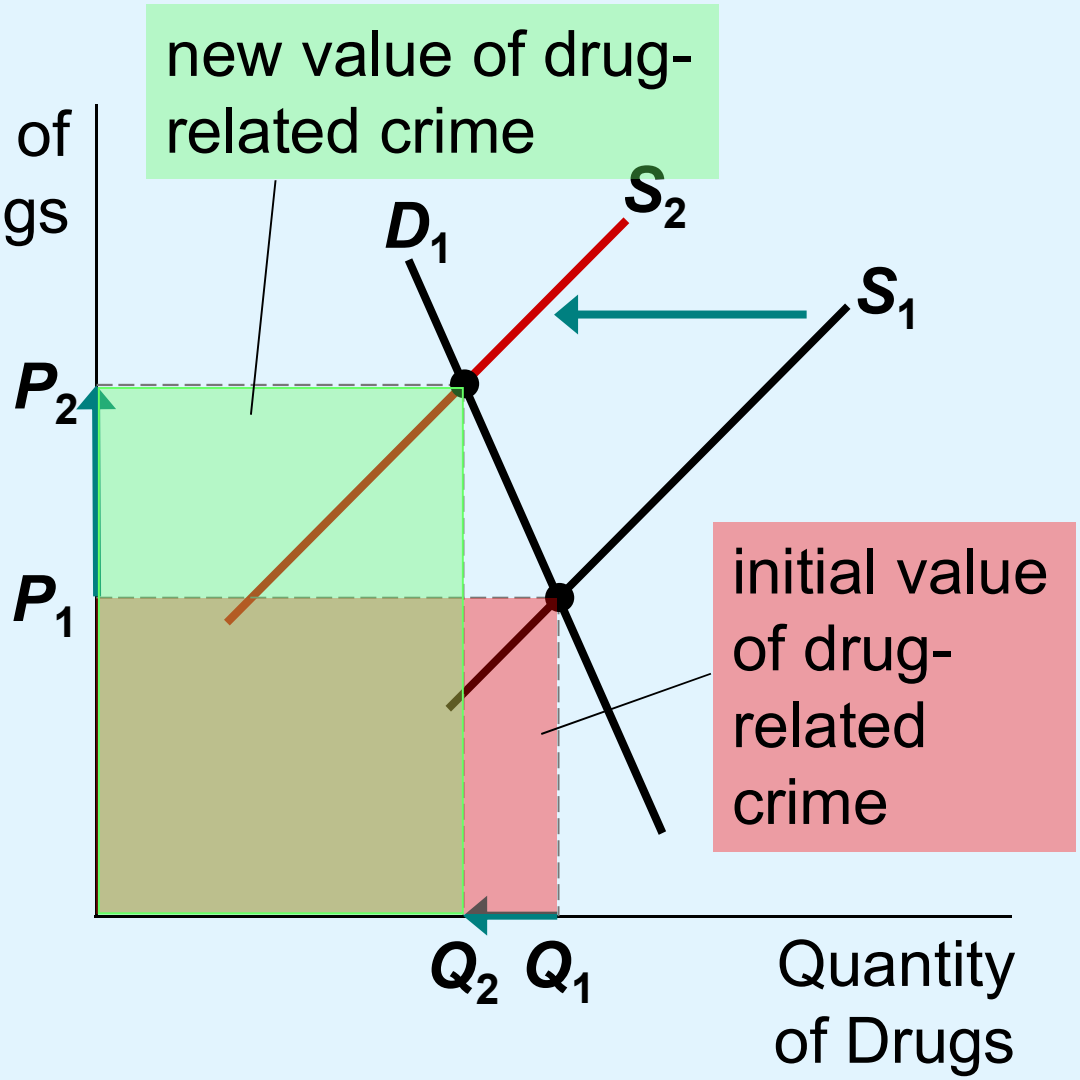


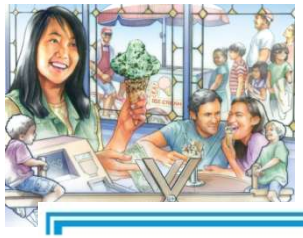
Policy 1: Interdiction

Interdiction reduces the supply of drugs.

Demand for drugs is inelastic: ***P*** rises proportionally more than ***Q*** falls.

Result: an increase in total spending on drugs, and in drug-related crime





Does Drug Interdiction Increase or Decrease Drug-related Crime?

2. Policy of drug education

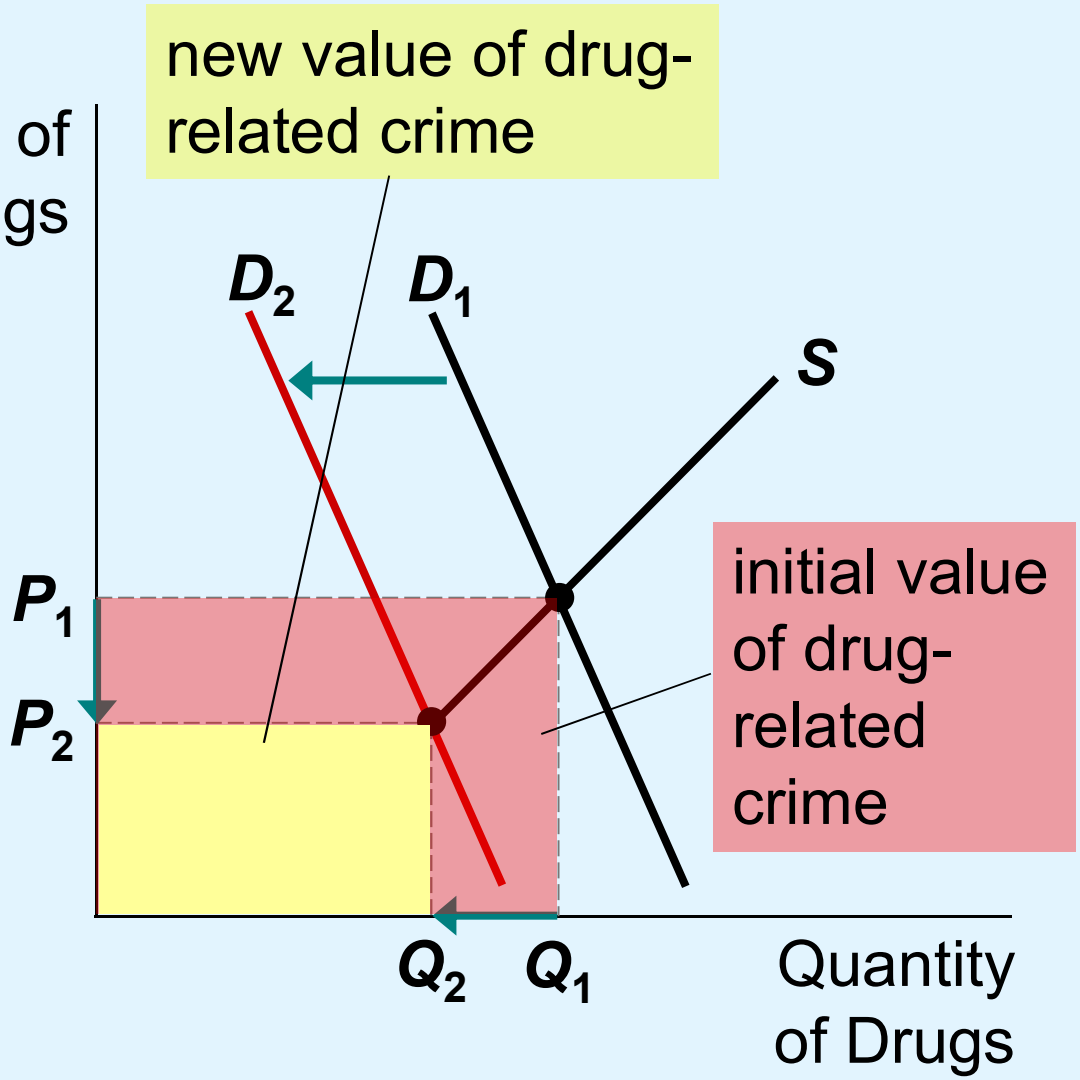
- Reduce demand for illegal drugs
- Left shift of demand curve
- Lower quantity
- Lower price
- Reduce drug-related crime

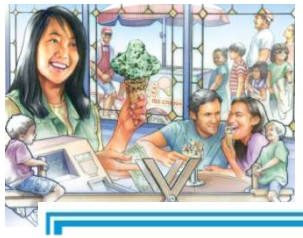
Policy 2: Education

Education reduces the demand for drugs.

P and ***Q*** fall.

Result:
A decrease in total spending on drugs, and in drug-related crime.



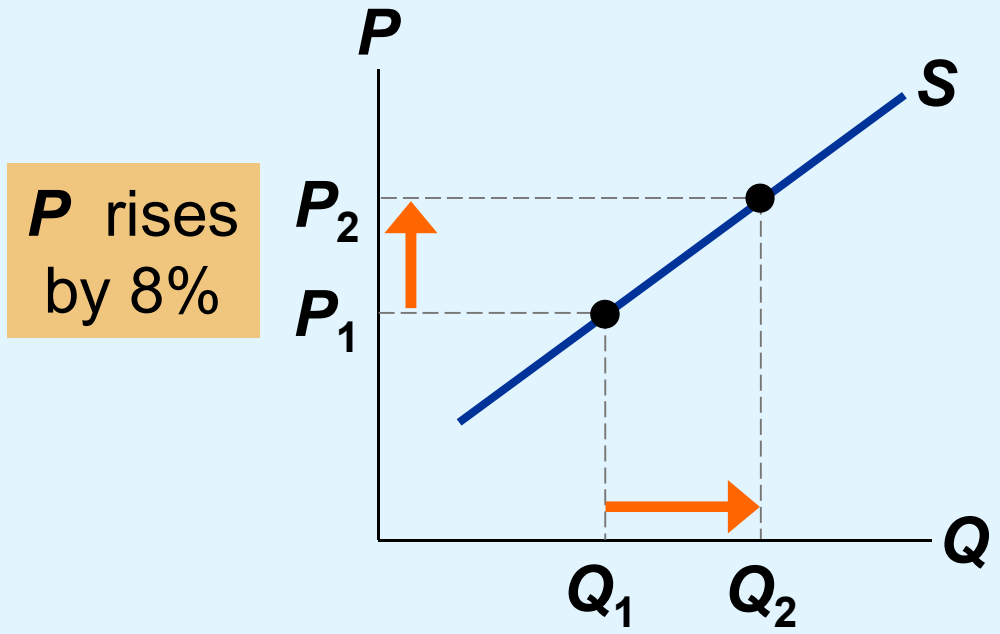


The Price Elasticity of Supply

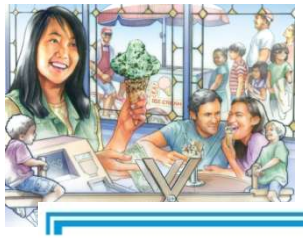
- Price elasticity of supply
  - How much the quantity supplied of a good responds to a change in the price of that good
  - Percentage change in quantity supplied
    - Divided by the percentage change in price
  - Loosely speaking, it measures sellers' price-sensitivity

Price Elasticity of Supply

$$\text{Price elasticity of supply} = \frac{\text{percentage change in } Q^s}{\text{percentage change in P}} = \frac{16\%}{8\%} = 2$$

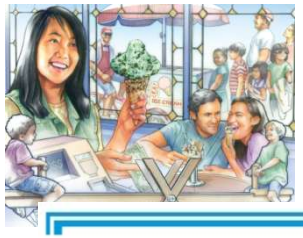


Again, we use the midpoint method to compute the percentage changes.



The Price Elasticity of Supply

- Variety of supply curves
  - Supply is unit elastic
    - Price elasticity of supply = 1
  - Supply is elastic
    - Price elasticity of supply > 1
  - Supply is inelastic
    - Price elasticity of supply < 1



The Price Elasticity of Supply

- Variety of supply curves
  - Supply is perfectly inelastic
    - Price elasticity of supply = 0
    - Supply curve is vertical
  - Supply is perfectly elastic
    - Price elasticity of supply = infinity
    - Supply curve is horizontal
- The flatter the supply curve
  - The greater the price elasticity of supply

Perfectly inelastic supply

$$\text{Price elasticity of supply} = \frac{\%\ \text{change in } \boldsymbol{Q}}{\%\ \text{change in } \boldsymbol{P}} = \frac{0\%}{10\%} = 0$$

S curve:

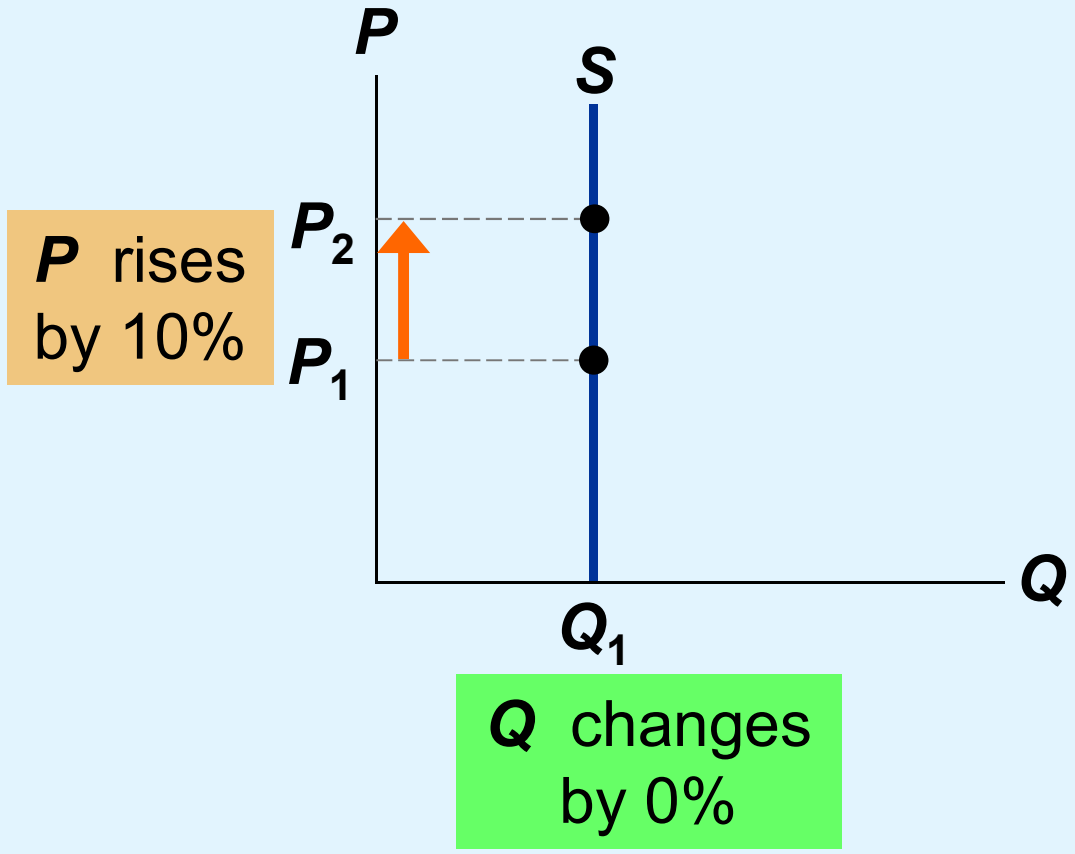
vertical

Sellers' price sensitivity:

none

Elasticity:

0



Inelastic supply

$$\text{Price elasticity of supply} = \frac{\%\text{ change in } \boldsymbol{Q}}{\%\text{ change in } \boldsymbol{P}} = \frac{< 10\%}{10\%} < 1$$

S curve:

relatively steep

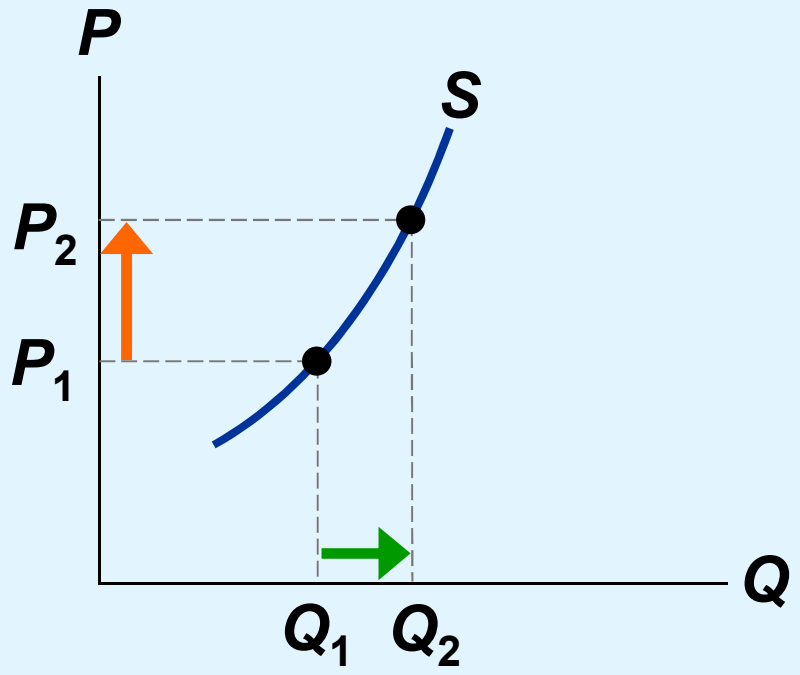
Sellers' price sensitivity:

relatively low

Elasticity:

< 1

P rises by 10%



Q rises less than 10%

Unit elastic supply

$$\text{Price elasticity of supply} = \frac{\%\text{ change in } \boldsymbol{Q}}{\%\text{ change in } \boldsymbol{P}} = \frac{10\%}{10\%} = 1$$

S curve:

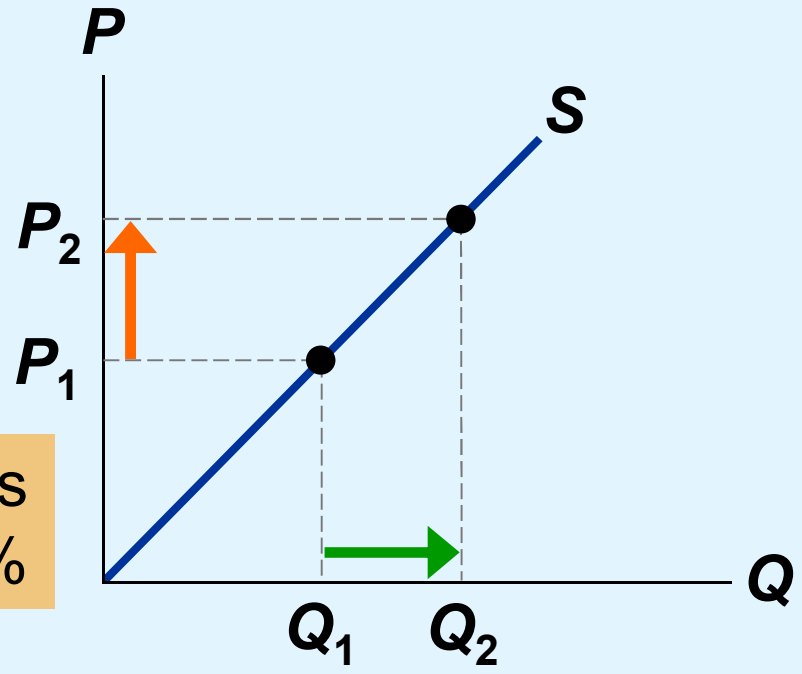
intermediate slope

Sellers' price sensitivity:

intermediate

Elasticity:

= 1



P rises by 10%

Q rises by 10%

Elastic supply

$$\text{Price elasticity of supply} = \frac{\%\text{ change in } \boldsymbol{Q}}{\%\text{ change in } \boldsymbol{P}} = \frac{> 10\%}{10\%} > 1$$

S curve:

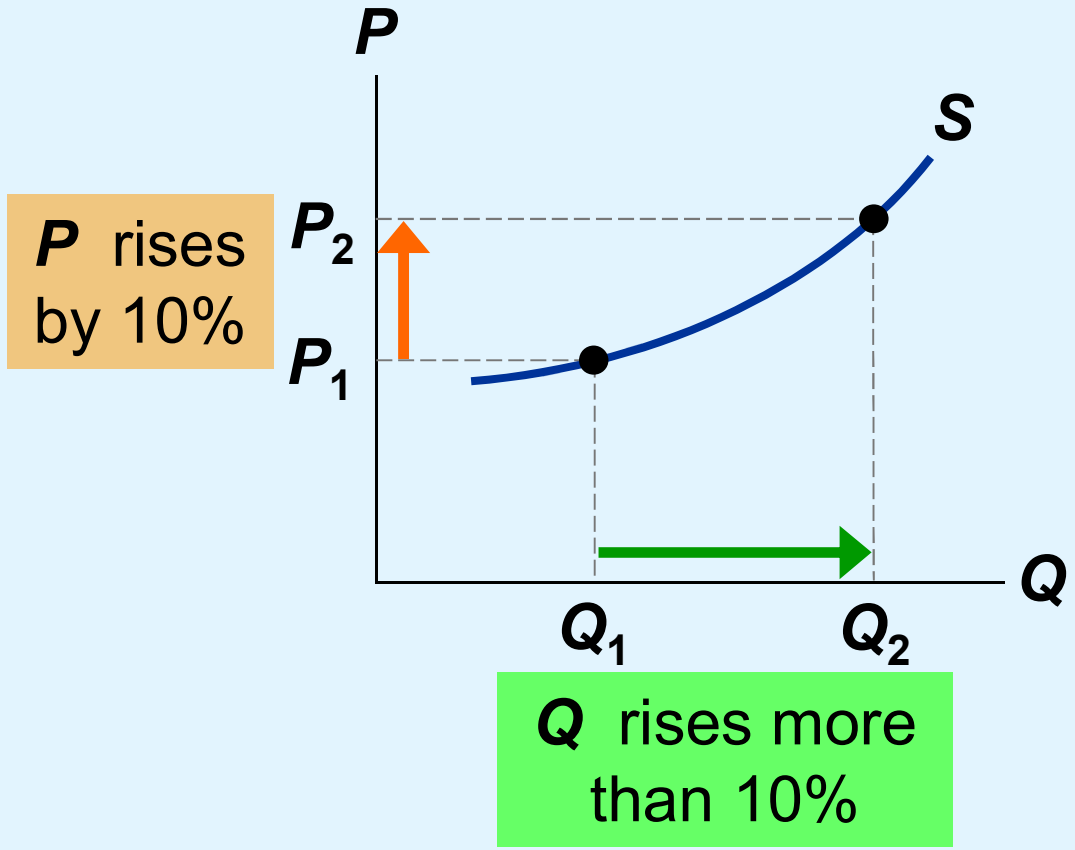
relatively flat

Sellers' price sensitivity:

relatively high

Elasticity:

> 1



Perfectly elastic supply

$$\text{Price elasticity of supply} = \frac{\%\ \text{change in } \boldsymbol{Q}}{\%\ \text{change in } \boldsymbol{P}} = \frac{\text{any } \%}{0\%} = \text{infinity}$$

S curve:

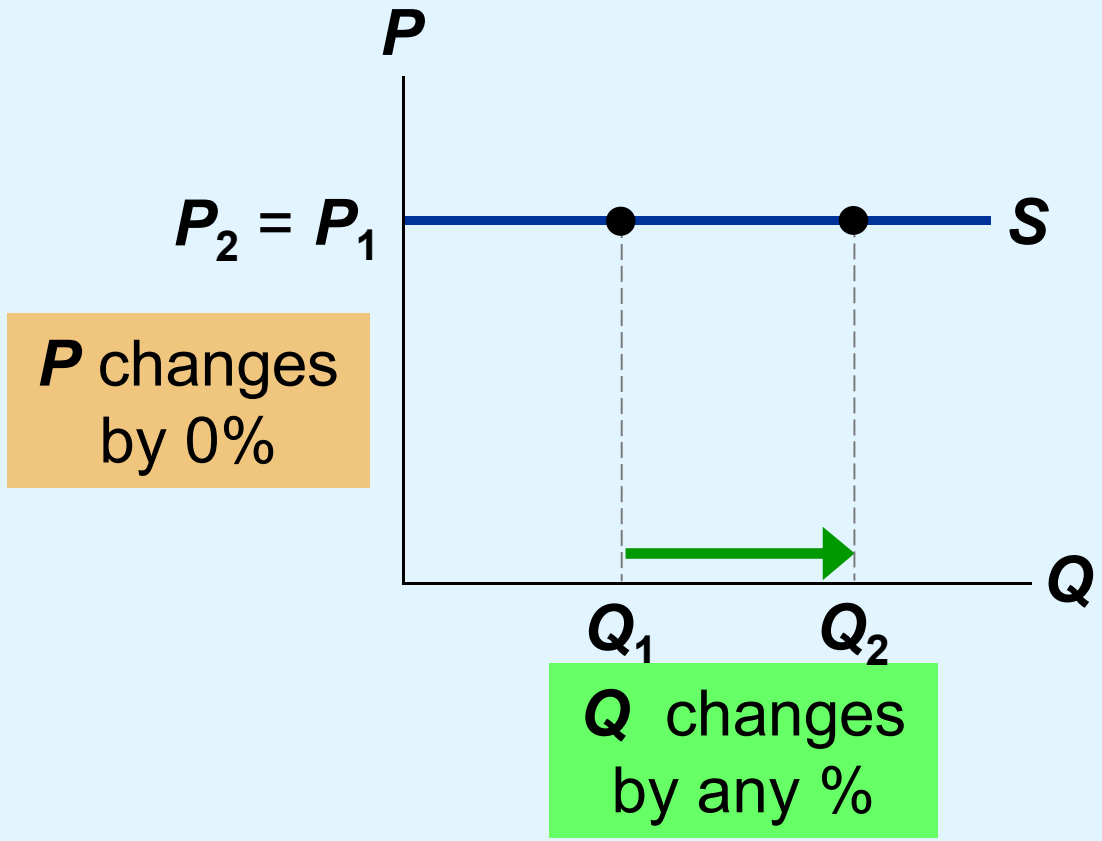
horizontal

Sellers' price sensitivity:

extreme

Elasticity:

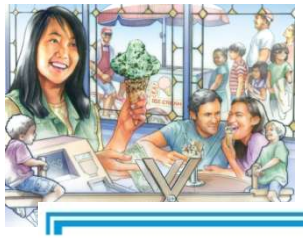
infinity



Question

- The price of coffee rose sharply last month, while the quantity sold remained the same. Five people suggest various explanations
  - Leonard: Demand increased, but supply was perfectly inelastic
  - Sheldon: Demand increased, but it was perfectly inelastic
  - Penny: Demand increased, but supply decreased at the same time
  - Howard: Supply decreased, but demand was unit elastic
  - Raj: Supply decreased, but demand was perfectly inelastic

Who could *possibly* right?



The Determinants of Supply Elasticity

- Greater price elasticity of supply
  - The more easily sellers can change the quantity they produce
    - Supply of beachfront property - harder to vary and thus less elastic than supply of *new cars*
- Price elasticity of supply is greater in the long run than in the short run
  - In the long run: firms can build new factories, or new firms may be able to enter the market

Active Learning 3

Elasticity and changes in equilibrium

The supply of beachfront property is inelastic.
The supply of new cars is elastic.

Suppose population growth causes demand for both goods to double (at each price, Q^d doubles).

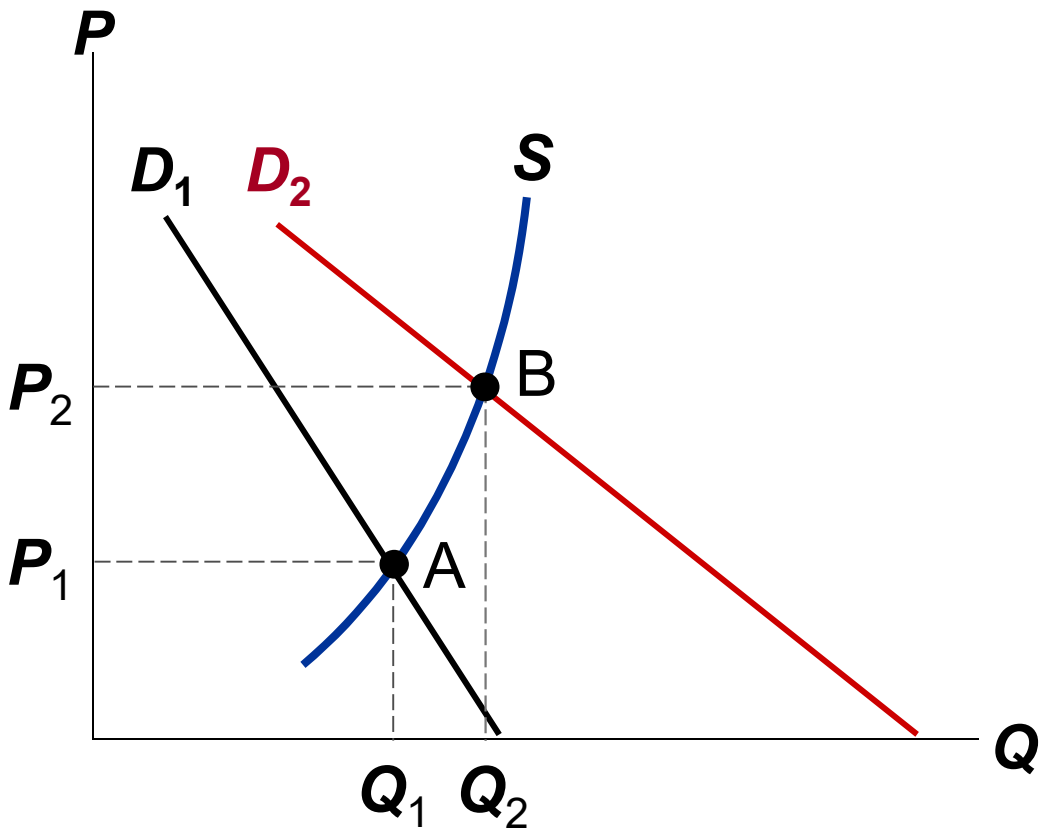
- For which product will ***P*** change the most?
- For which product will ***Q*** change the most?

Active Learning 3

Answers

When supply is *inelastic*, an increase in demand has a bigger impact on price than on quantity.

Beachfront property (inelastic supply):

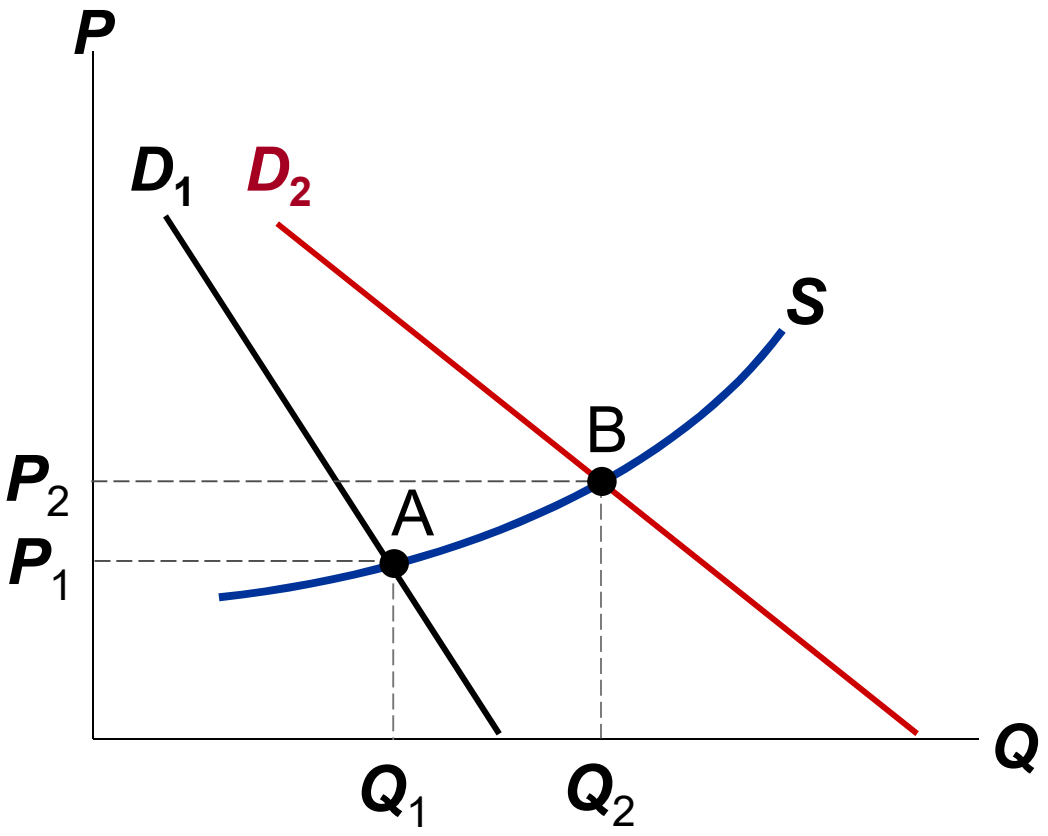


Active Learning 3

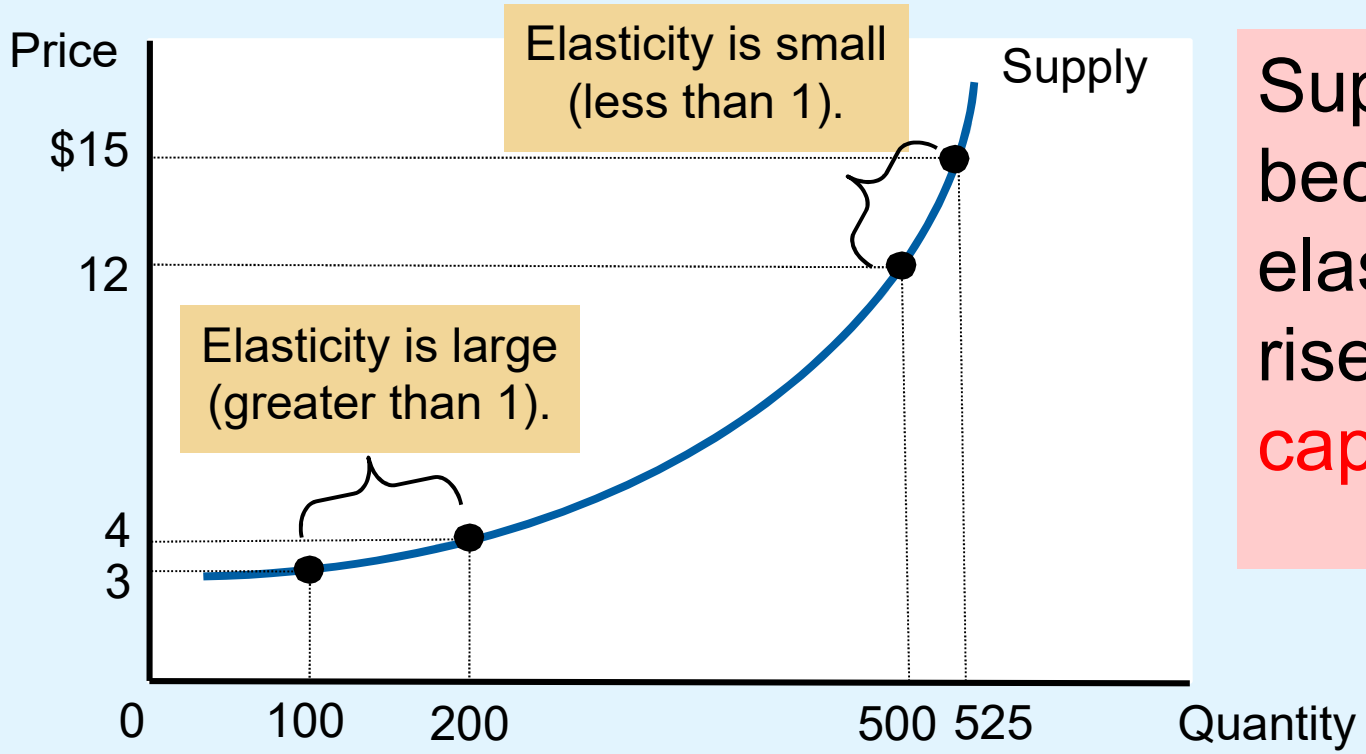
Answers

When supply is *elastic*, an increase in demand has a bigger impact on quantity than on price.

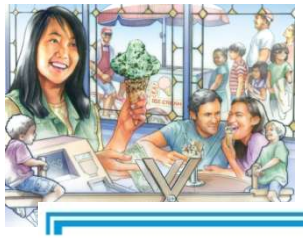
New cars (elastic supply):



How the Price Elasticity of Supply Can Vary



Supply often becomes less elastic as Q rises, due to capacity limits.

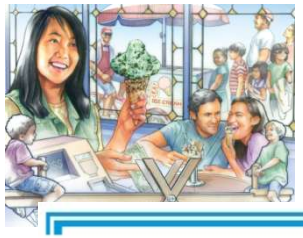


Other Elasticities of Demand

- Income elasticity of demand
  - How much the quantity demanded of a good responds to a change in consumers' income
  - Percentage change in quantity demanded
    - Divided by the percentage change in income
  - Normal goods: income elasticity > 0

    收入增加,需求增加
  - Inferior goods: income elasticity < 0

    收入增加,需求减少

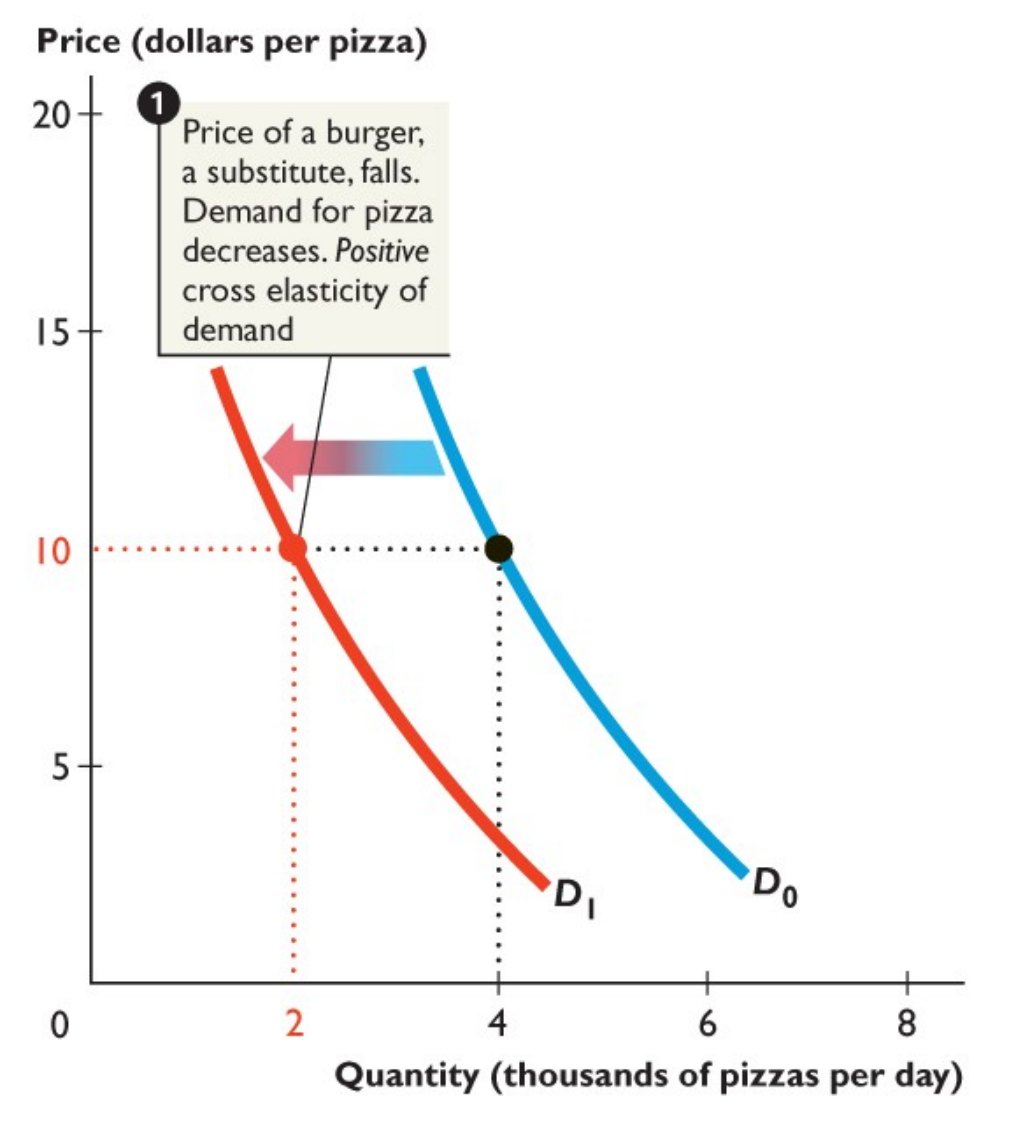


Other Elasticities of Demand

- Cross-price elasticity of demand
  - How much the Q^d of one good responds to a change in the price of another good
  - Percentage change in Q^d of the first good
    - Divided by the percentage change in price of the second good
  - Substitutes: cross-price elasticity > 0
  - Complements: cross-price elasticity < 0

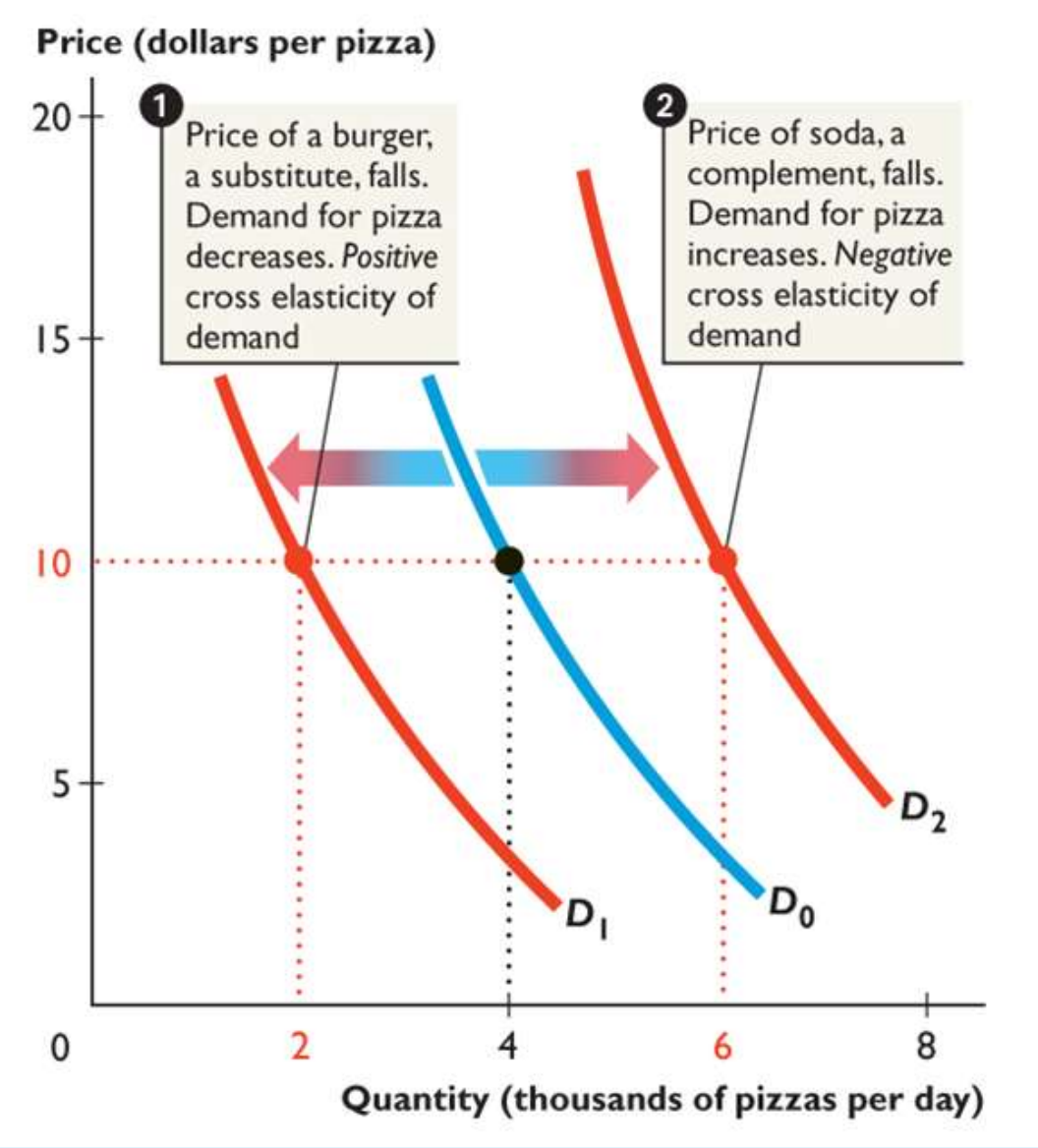
需求的交叉彈性→A產品的數量變化對應到B產品的價格改變

Cross-price elasticity of demand

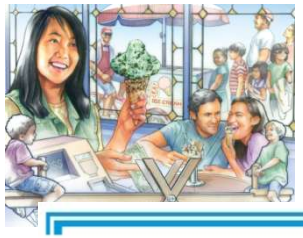


1. Pizzas and burgers are substitutes.
2. When the price of a burger falls, the demand for pizza decreases.
3. Cross elasticity of demand is positive.

Cross-price elasticity of demand



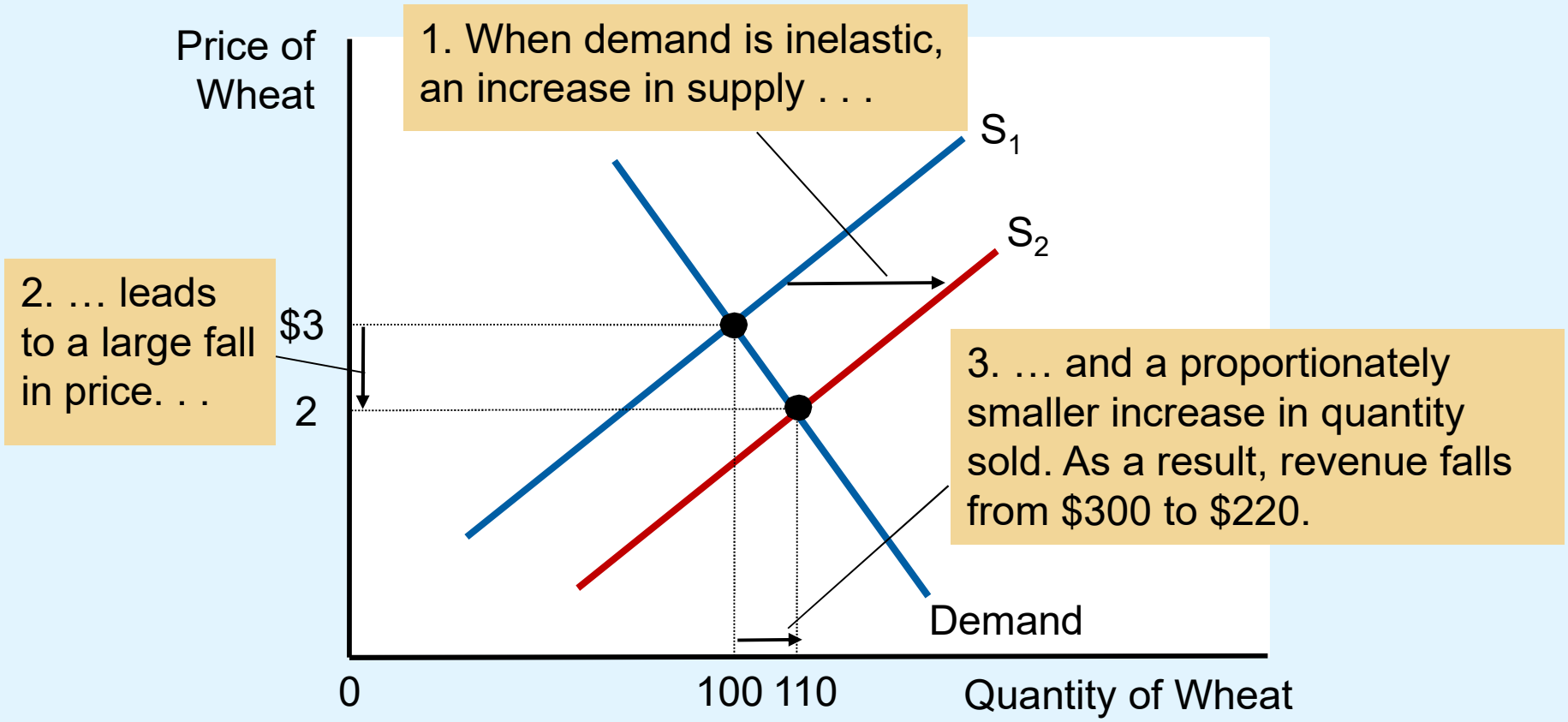
1. Pizzas and soda are complements.
2. When the price of soda falls, the demand for pizza increases.
3. Cross elasticity of demand is negative.

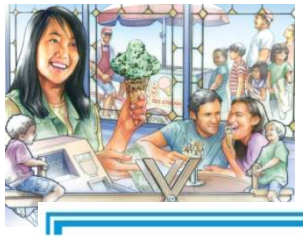


Applications

- Can Good News for Farming Be Bad News for Farmers?
  - New hybrid of wheat – increase production per acre 20%
    - Supply curve shifts to the right
    - Higher quantity and lower price
    - Demand is inelastic: total revenue falls
  - Paradox of public policy: induce farmers not to plant crops

An Increase in Supply in the Market for Wheat

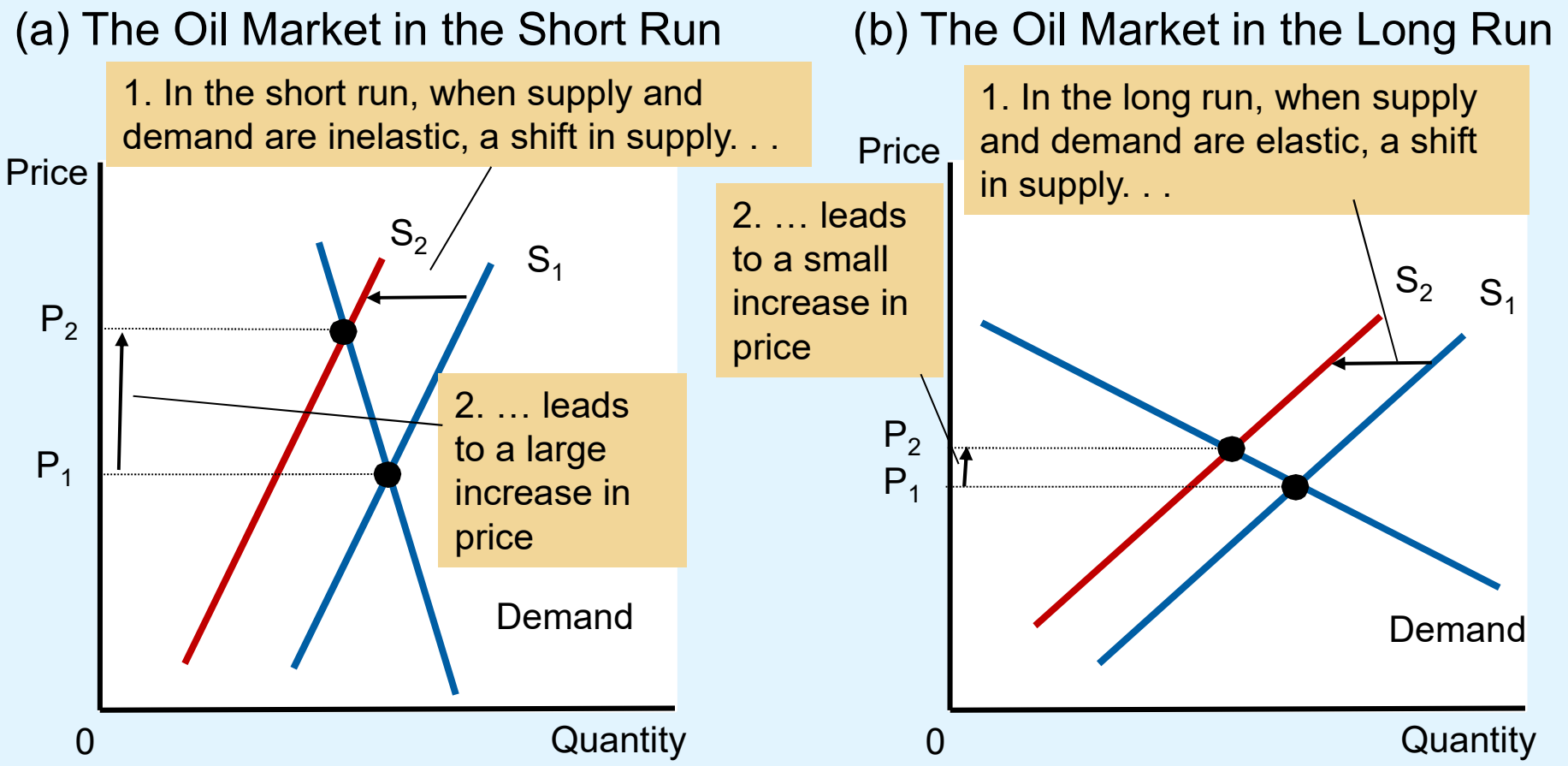




Applications

- Why Did OPEC Fail to Keep the Price of Oil High?
  - Increase in prices 1973-1974, 1971-1981
  - Short-run: supply and demand are inelastic
    - Decrease in supply: large increase in price
  - Long-run: supply and demand are elastic
    - Decrease in supply: small increase in price

A Reduction in Supply in the World Market for Oil



長期情況下可能會讓頁岩油業者加入市場

Summary

- Elasticity measures the responsiveness of Q^d or Q^s to one of its determinants.
- Price elasticity of demand equals percentage change in Q^d divided by percentage change in P.
  When it's less than one, demand is "inelastic."
  When greater than one, demand is "elastic."
- When demand is inelastic, total revenue rises when price rises. When demand is elastic, total revenue falls when price rises.

Summary

- Demand is less elastic in the short run, for necessities, for broadly defined goods, and for goods with few close substitutes.
- Price elasticity of supply equals percentage change in Q^s divided by percentage change in P.
  When it's less than one, supply is "inelastic."
  When greater than one, supply is "elastic."
- Price elasticity of supply is greater in the long run than in the short run.

Summary

- The income elasticity of demand measures how much quantity demanded responds to changes in buyers' incomes.
- The cross-price elasticity of demand measures how much demand for one good responds to changes in the price of another good.
- The tools of supply and demand can be applied in many different kinds of markets. This chapter uses them to analyze the market for wheat, the market for oil, and the market for illegal drugs.

Elasticity, Price Change, and Expenditure

If demands is . . .	A price increase will . . .	A price reduction will . . .
elastic ($\epsilon > 1$)	reduce total expenditure	increase total expenditure
inelastic ($\epsilon < 1$)	increase total expenditure	reduce total expenditure





What do You do When Starbucks Raises the Price of a Latte?



Dunkin' Donuts





Eye on Elasticity at the Coffee Shop (1 of 3)

What Do You Do When Starbucks Raises the Price of a Latte?

If you are like most people, you complain when the price of a latte rises, but the first thing to do is look at the prices of a latte in other nearby coffee shops.

Suppose that their prices have not increased and you can get an acceptable latte for a lower price at Dunkin' Doughnuts.

You decide to switch to Dunkin' Doughnuts and keep drinking the same quantity of latte.

But soon though, all the coffee shops in your neighbourhood have raised their price. Your daily latte is more expensive, so you go back to Starbucks for your latte.

Eye on Elasticity at the Coffee Shop (2 of 3)

What Do You Do When Starbucks Raises the Price of a Latte?

Price Elasticity of Demand

We can translate your possible response to a hike in Starbucks' price into estimates of your price elasticity of demand for **Starbucks latte** and for **latte in general**.

Close Substitutes

A Starbucks latte and a Dunkin' Doughnuts latte are different, but for some people they are reasonably close substitutes.

So when Starbucks alone raises its price, the quantity sold by Starbucks changes a lot—**the demand for a Starbucks latte is elastic**.

Eye on Elasticity at the Coffee Shop (3 of 3)

What Do You Do When Starbucks Raises the Price of a Latte?

Poor Substitutes

Later, when all the coffee shops have raised the price of a latte, you return to Starbucks but cut back a bit on your consumption of latte. Your price elasticity of demand for latte of all types is low—it is inelastic.

Latte, tea, and other drinks are poor substitutes.

So when the price of a latte rises, the quantity of latte bought decreases but not by very much—**the demand for latte of all types is inelastic**

Question

Consider public policy aimed at smoking.

a. Studies indicate that the price elasticity of demand for cigarettes is about 0.4. If a pack of cigarettes currently costs $5 and the government wants to reduce smoking by 20%, by how much should it increase the price?
b. If the government permanently increases the price of cigarettes, will the policy have a larger effect on smoking 1 year from now or 5 years from now?
c. Studies also find that teenagers have a higher price elasticity of demand than adults . Why might this be true?